



LegalEase

AI-Powered Legal Intelligence Platform

README / Project Documentation

A browser-based legal document analyzer, contract review assistant, and AI legal chatbot powered by Google Gemini when an API key is supplied.

Frontend	HTML5 + CSS3 + Vanilla JavaScript
AI Integration	Google Gemini 2.0 Flash REST API
Demo Mode	Built-in sample analysis and chatbot responses
Local Server	PowerShell HttpListener on localhost:4000

Source basis: the supplied index.html, style.css, app.js, server.ps1, and LegalEase logo.

1. Project Overview

LegalEase is a single-page web application designed to help users understand legal documents using an interactive frontend and optional Gemini AI integration. The supplied implementation includes document analysis, contract review, a legal AI chatbot, pricing UI, responsive navigation, animations, and a local PowerShell static server.

The HTML defines the main navigation and application sections, including Home, Features, How It Works, Analyze, AI Chat, and Pricing.

```
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```

Core capabilities

- **AI Document Analyzer:** paste legal text, select a sample, optionally provide a Gemini API key, and receive a structured risk analysis.
- **Contract Review:** review a contract with a selected focus: general overview, risks, obligations/deadlines, financial terms, or termination.
- **Legal AI Chat:** ask general legal questions, use quick prompts, and maintain an in-page chat history.
- **Demo mode:** the application can operate without an API key using built-in demo analysis and canned legal-topic responses.
- **Responsive UI:** mobile navigation, animated cards, smooth scrolling, intersection-observer reveal animations, and interactive toast messages.

Important implementation note

The supplied frontend calls the Gemini REST endpoint directly from browser JavaScript and accepts the API key in a password field. This is suitable for a local/demo implementation, but a production deployment should normally keep the API key on a server-side backend rather than exposing it to browser code.

2. Technology Stack

Layer	Technology / Implementation
Markup	HTML5 single-page application
Styling	CSS3 with custom variables, responsive layouts, glass-style cards, gradients and animations
Client logic	Vanilla JavaScript
AI	Google Gemini 2.0 Flash REST API
Fonts	Inter and Playfair Display via Google Fonts
Local server	PowerShell + .NET HttpListener
Static assets	HTML, CSS, JS and JPG logo

3. Application Features

Document Analysis

The analyzer accepts pasted legal text and includes Sample NDA, Sample Contract, and Clear controls. The UI also provides a Gemini API key field and an Analyze with AI action. ■filecite■turn0file1■L243-L275■

Risk Analysis Output

AI analysis is expected to return document type, an overall 0–100 risk score, risk level, executive summary, key risks, key clauses, and recommendations. The JavaScript renders these results into cards. ■filecite■turn0file0■L185-L216■

Contract Review

The contract reviewer supports five review focuses: general overview, risk identification, obligations/deadlines, financial terms, and termination conditions. ■filecite■turn0file1■L304-L340■

Contract Results

The review output displays contract type, parties, assessment, executive summary, financial terms, key obligations, critical dates, and negotiation points. ■filecite■turn0file0■L436-L510■

AI Chatbot

The chat interface supports quick questions about NDAs, liability/indemnification, non-competes, service agreements, force majeure, and payment terms. ■filecite■turn0file1■L359-L416■

Demo Mode

Without a Gemini key, document analysis and chat use local JavaScript fallback logic. The app explicitly displays a demo-mode notification so the user knows that real AI analysis requires a key. ■filecite■turn0file0■L207-L216■

4. User Interface Sections

Section	Purpose
Home / Hero	Introduces LegalEase and provides Analyze a Document and Review a Contract actions.
Features	Presents Smart Document Analysis, Contract Review, Legal Risk Assessment and Real-time Collaboration.
How It Works	Explains upload/paste → AI analysis → clear insights workflow.
Analyze	Interactive legal document analysis workspace.
Review	Interactive contract summarization/review workspace.
AI Chat	Conversational legal assistant with quick prompts.
Pricing	Starter, Professional and Enterprise pricing cards.

5. Project Structure

```
LegalEase/
├── index.html
├── style.css
├── app.js
├── server.ps1
└── legalease_logo.jpg
```

index.html — application structure, navigation, hero, feature cards, analyzer, contract review, chatbot, pricing and footer. The document loads style.css and app.js. [filecite\turn0file1\L1-L12](#)

style.css — visual design system and responsive layout. The supplied theme uses navy, gold, blue, purple, green and red variables, with glass-like surfaces and animated UI elements. [filecite\turn0file2\L5-L33](#)

app.js — application behavior: navigation, sample data, Gemini requests, document analysis, contract review, chat, toast notifications and animations.

server.ps1 — lightweight local static file server. It listens on localhost:4000 and serves HTML, CSS, JS, image and font MIME types. [filecite\turn0file3\L1-L29](#)

legalease_logo.jpg — supplied LegalEase logo used by the navigation and footer.

6. Application Flow

Step	Flow
1	User opens the LegalEase page through the local server.
2	User navigates to Analyze, Review, or AI Chat.
3	User pastes legal content or loads a built-in sample.
4	If a Gemini API key is present, browser JavaScript sends a request to Gemini 2.0 Flash.
5	The response is parsed/rendered into structured UI cards.
6	If no key is present, the corresponding demo fallback is used.

Gemini request details

The supplied JavaScript targets the Gemini 2.0 Flash generateContent endpoint and sends a text prompt inside the request body. The configured generation settings use temperature 0.3 and a maximum of 2048 output tokens. [filecite\turn0file0\L120-L142](#)

7. VS Code Setup & Installation

The supplied project is intentionally lightweight: it does not contain a package.json or Node.js dependency setup. The main requirements are a modern browser and, for the supplied server script, Windows PowerShell.

Step 1 — Create/open the project folder

Place these files in one folder: **index.html**, **style.css**, **app.js**, **server.ps1**, and **legalease_logo.jpg**.

Step 2 — Open in VS Code

File → Open Folder → select your LegalEase project folder

Confirm that the HTML references match the filenames exactly: **style.css**, **app.js**, and **legalease_logo.jpg**.

Step 3 — Configure the PowerShell server path

The supplied server script currently contains a Windows-specific root path. Change **\$rootPath** to the actual folder location on your computer before running it. The server uses this path to locate the static files.

■filecite■turn0file3■L1-L6■

```
$port = 4000
$rootPath = "C:\Path\To\Your\LegalEase"
```

Step 4 — Start the local server

```
PowerShell
cd C:\Path\To\Your\LegalEase
Set-ExecutionPolicy -Scope Process -ExecutionPolicy Bypass
.\server.ps1
```

The script reports that LegalEase is running at **http://localhost:4000**. ■filecite■turn0file3■L8-L14■

Step 5 — Open the application

<http://localhost:4000>

The server maps the root request / to index.html. ■filecite■turn0file3■L31-L40■

Step 6 — Optional Gemini API key

For real AI analysis, enter a valid Gemini API key in the analyzer or chat interface. The key is read by the browser-side JavaScript and used in the Gemini request.

8. How to Use LegalEase

A. Analyze a legal document

- Open **Analyze** from the navigation.
- Paste legal text into Document Input, or choose Sample NDA / Sample Contract.
- Optionally enter the Gemini API key.
- Click **Analyze with AI**.
- Review document type, risk score, risk level, executive summary, key risks, key clauses and recommendations.

The analyzer shows a loading sequence for parsing structure, identifying clauses, assessing risks and generating a summary. `filecite turn0file1 L277-L297`

B. Review a contract

- Open **Review**.
- Paste the contract or use Load Sample.
- Choose the desired Review Focus.
- Click **Review Contract**.
- Read the contract overview, parties, assessment, financial terms, obligations, critical dates and negotiation points.

C. Use the AI chatbot

- Open **AI Chat**.
- Choose a Quick Question or type your own legal question.
- Press Enter to send (Shift+Enter can be used for a new line).
- Use Clear Chat to reset the conversation.
- Add a Gemini API key for real AI answers; otherwise the built-in demo responses are used.

The chatbot code keeps recent conversation messages, sends the recent context to Gemini when a key is present, and otherwise uses predefined topic responses. `filecite turn0file0 L638-L672`

9. Pricing UI Included in the Supplied Frontend

The pricing section is currently presentation/UI only. The supplied buttons display a 'Signup flow coming soon!' toast rather than implementing account creation or billing. `filecite turn0file0 L567-L573`

Plan	Displayed price	Included highlights
Starter	\$0/month	10 analyses/month, 5 contract reviews/month, basic risk scoring
Professional	\$49/month	Unlimited analyses/reviews, advanced scoring, 5-user collaboration, PDF/Word export
Enterprise	\$199/month	Unlimited members, custom AI fine-tuning, API access, dedicated support

10. Troubleshooting

Page does not load

Check that PowerShell shows the listener started and open `http://localhost:4000`. Verify the `$rootPath` points to the folder containing `index.html`.

404 for CSS/JS/logo

Make sure `style.css`, `app.js` and `legalease_logo.jpg` are in the same project directory as `index.html`. The server only serves files that exist under its configured root.

Gemini API error

Check the API key, browser network errors, quota/permissions, and the Gemini endpoint/model availability. The JavaScript displays the returned API error message when the request is not successful.

AI analysis returns malformed output

The application expects JSON for analysis/review and attempts to extract a JSON object from the model response. If the model returns incompatible content, parsing can fail.

Demo works but real AI does not

This is expected when no key is supplied: the app has explicit fallback demo logic. Add the API key only when you want the Gemini-backed path.

PowerShell execution policy blocks script

Run `Set-ExecutionPolicy -Scope Process -ExecutionPolicy Bypass` in the current PowerShell session, then start `.\server.ps1`.

11. Security & Production Considerations

- **Protect API keys:** the supplied frontend accepts a Gemini key directly in the browser. For production, place the Gemini call behind a secure backend API.
- **Validate user input:** legal documents can contain arbitrary text. Add server-side validation, size limits, authentication and rate limiting before exposing the service publicly.
- **Protect sensitive legal documents:** add an explicit data-retention policy, access controls, encryption, audit logging and secure storage if uploaded documents are persisted.
- **Do not treat AI output as legal advice:** the UI itself states that AI responses are informational and recommends consulting a qualified attorney. ■filecite■turn0file1■L378-L381■
- **Verify marketing claims before publishing:** the UI contains claims such as accuracy, number of documents, certifications and firms served. These are presentation text in the supplied source and should be independently substantiated before being used as real-world claims.

12. Current Scope vs. Future Backend Work

Included in supplied code	Not implemented as a working backend
Single-page responsive UI	User authentication / real login
Gemini browser API calls	Secure server-side API proxy

Demo analysis and chat fallbacks	Persistent document database
Local PowerShell static server	Real subscription/billing integration
Pricing and signup UI	Real collaboration/workspaces
Legal analysis result rendering	Production document upload/storage pipeline

13. Quick Reference

Project files

index.html

→ UI structure

style.css

→ visual design and responsive styles

app.js

→ AI, analyzer, reviewer, chat and interactions

server.ps1

→ local static server on port 4000

legalease_logo.jpg

→ LegalEase brand logo

Main JavaScript functions

Function	Purpose
analyzeDocument()	Runs document analysis using Gemini or demo fallback.
reviewContract()	Runs contract review using Gemini or demo fallback.
sendChatMessage()	Sends chatbot messages using Gemini or demo responses.
loadSample()	Loads sample NDA or contract text into the analyzer.
renderAnalysis()	Renders structured document-analysis results.
renderReview()	Renders structured contract-review results.
showToast()	Displays temporary status messages.

Quick start

1. Open the LegalEase folder in VS Code

2. Edit \$rootPath in server.ps1

3. Run: .\server.ps1

4. Open: http://localhost:4000

5. Try Sample NDA → Analyze with AI

6. Try Review → Load Sample → Review Contract

7. Try AI Chat → ask a legal question

This README describes the supplied project as provided. It does not claim that unimplemented backend, billing, authentication, database, collaboration, or production-security functionality is already present.

End of README